

WORK PROCEDURE

PRO-OP-ITH-006A

STANDARD FOR V-30 CONVENTIONAL BORING

OBJECTIVE

The objective of this procedure is to define safe working methods for conventional boring with the drilling head V-30.

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

INTERPRETATION

Personnel with the authority for the interpretation of this document are the director general and operations director. Any request for amendment, revision or update must be made with one of these persons.

RESPONSIBILITIES

The worker responsible for the drilling must ensure to follow these steps and follow the instruction that come from the engineering department.

METHOD

- 1) MRI employees are allowed to move the drilling equipment into the workplace using the company's equipment provided all company regulations and directives are to be followed respecting the company's policies.
- 2) A visual inspection of the unit is completed once the unit arrives at the workplace. A visual inspection is also completed of ventilation, power supply and air and water distribution.
- 3) The electric cable is unloaded, inspected and hooked up to the ground fault distribution box. (a company electrician is required just in case a motor rotation re-connect is required)

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- 4) Verification of all control levers is required prior to moving the unit. They must be in the neutral position before powering up the unit.
- 5) A visual inspection of the unit is required for oil levels, gauges, leaks, etc. prior to moving the unit.

INSPECTIONS

It is essential to carry out a rigorous inspection of our equipment as soon as we have to operate it. In the case of CUBEX and boosters, particular attention must be paid to the high-temperature (800 degrees F) braided hoses of the compressors. For safety reasons, only the type of hose as shown below should be used.



- 6) Before the unit is moved to the first line to be drilled, with respect to the plans and specifications, a wash down of the area, a check scale and an inspection for explosives is performed in the drilling area.
- 7) When the drill is moved in the drilling area, we take precautions to avoid damage to the cable connected to the drill and check if there are no leaks or abnormalities of the drill in motion. If the extension should be handled when it

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is turned on, you must use the plastic hook. Barricades and or signs are installed to advise all other workers around of the ongoing operations (Drilling, Open Holes, etc.). A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.).

- 8) After reaching the drilling point, the drill is turned off with the **E-STOP** and the electric cable is then hung to the wall at chest height (4 to 5 feet off the bench) to eliminate, as much as possible, any contact by loose or motor vehicles. Also, as the cable is being hung, a visual inspection of the cable is done
- 9) The connection of the low-pressure air hose (2 ½ in. dia.) from the mine air supply to the drill's booster compressor is the next priority. The airline is blown clean and safety-restraining devices (whip checks) are installed at each connection and splice (if there happens to be one). At this point, an inspection of the retaining device is important. You must check the condition of the wire and the proper functioning of clamping mechanisms



The use of the E-Stop to stop the drill will result of cutting the 600 V current on all extensions.

- 10) The re-enforced high-pressure hose (2 ½) is inspected, blown and connected from the compressor to the drill unit using whip checks at all the connections. (Note: Stem nuts on the 2 ½ in. dia. hoses are designed so that they cannot be interchanged between high and low pressure so as to eliminate, as much as possible, any chance of a potential accident.)

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Special attention required

Connections and hoses should be thoroughly inspected to ensure that the collars are tight, that the threads of the couplings are in good condition, and that the hoses are not faulty.

Difference in thickness between the two connections. This detail creates an incompatibility between the two types of fittings

Locking system integrated into the fittings. A check for each use is required. This mechanism does not replace the metal pin



- 11) The 1 in. dia. water hose is inspected, blown and connected from the mine supply to the drill.
- 12) All hoses are hung under the electric cable to eliminate any contact with loose or mine equipment.
- 13) If it is not physically possible to hang the cable and hoses (ex. due to stope dimensions) they are laid out in such a fashion to eliminate contact or crushing. They may be raised off the bench using drums or wooden tri-pods and marked (painted or identified with flagging) so that they will be visible.
- 14) Air and water lines are then slowly energized and inspected for leaks.
- 15) The pilot hole of 6½ in. in dia. requires a helper to aid in the manipulation of the hammer and bit and rod changes due to the physical effort involved.
- 16) The drill unit is then positioned over the drilling target (on line and at the desired angle) as required in the plans and specifications supplied using a magnetic laser and a digital laser inclinometer for quality control
- 17) When moving the rotating mast, special attention must be paid to the hydraulic hoses, they could get stuck. If so, go back and move the mast to loosen them. If an effort is necessary to loosen the hoses, make sure you are in a stable position.

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- 18) MRI employees are trained to sharpen and re-face drill bits at the work site. Hand held sharpeners and grinders are used for this purpose and all personal protective equipment (PPE) is required to be used.
- 19) The drill consol must be positioned in such a way as to protect the employees from all rock projections, which may occur when starting to collar the pilot hole.
- 20) During the drilling of the hole no one (employee or other) is authorized to circulate in front of the drilling unit.
- 21) To remove or install drill rods the gear box must be raised and de-energized. The drill must be in the idle position before rods are manipulated.
- 22) Once the 6½ in. dia. pilot hole has been completed it must be reamed to 10½ in. in diameter for the V-30 guide sleeve. The same method for rod installation and removal applies. The hammer and reamer bit is removed at the break through level and the V-30 head is pinned to the rods.
- 23) **When a break through holes is done, it is important to push, WITHOUT ROTATION, 1 rod longer than the length of the break through;**
- 24) For production holes greater than 6½ inch in diameter, the operator will require the assistance of a driller helper when handling the hammer. The hammer must be handled using the transport carriage. **Make sure that the transport carriage is stable before laying the hammer on.** Bits changes and rods handling can be handled by the operator only, depending on the length of holes to be drilled. **We favor the presence of a second worker for drilling and tearing rods for holes 200 feet and over. If it is not possible to have this 2nd worker, the driller must provide breaks**
- 25) One of three methods can be used to install the V-30 head onto the rods in preparation to drill the 30 in. diameter hole.
 - a) Using a grade 80 chain (7100 lbs) and a scoop tram, the head is tied off and lifted into position. The head's guide is positioned to accept the drill rod, which has broken through at the back of the work area on the level below. The scoop is then used to adjust the alignment. The gearbox is then activated and the head is threaded onto the rod. The whole unit is then raised up to start reaming.
 - b) The more time consuming and less used procedure is using the same grade 80 chain and chain blocks to position the head. A high degree of attention and vigilance is required using this procedure because of the equipment used and the proximity of the employees to the head. The chain blocks must be installed to either lifting lugs or eyebolts. MRI personnel do not install these items, which are not part of our

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expertise. A scoop tram or lift truck is used to advance the head as close as possible to the installation point.

- 26) Signs and or barricades are installed at the entrance to the work area to advise every one of the work in progress and that there is an open hole situation progressing above. Only authorized personnel are allowed into the area.
- 27) Bit changes and a visual inspection is required periodically. The V-30 should be lowered and pulled out from the stope according with site procedure.
- 28) The head is then inspected for any possible repairs and the bits can then be changed.
- 29) Bit sharpening and re-facing can be done using the manual equipment (PPE's required) while the drilling continues.
- 30) Depending on the plans and specifications supplied, either a crown pillar is left in place, to protect workers, or the raise is broken through to the level and a steel plate is installed and identified.
- 31) When the raise is done the head is lowered to the lower level, the head is tied off to a scoop bucket or lift truck and removed from the work area. The rods are removed by raising them to the level above.
- 32) Once the 30 in. raise has been completed and the equipment has been moved out, a barricade or the necessary signs are posted so as all are notified that there is an open hole condition. The company is informed of the situation and they can undertake the necessary safety precautions required.
- 33) Procedure for dealing with rod failure:
 - a) Should a rod failure occur, immediately stop drilling, shut down electrical power and air supply to the drill.
 - b) A barricade or the necessary signs must be posted at the entrance to the work area in such a manner as to close off the work area and preventing all access to the site.
 - c) The driller must notify his work supervisor immediately and no work will resume until the supervisor is on site.
 - d) The driller assisted by the site supervisor will assess the situation and determine the means by which the failed rod or rods will be removed and replaced.
 - e) The site supervisor will remain on site throughout the recovery process.
 - f) All drill operation required to remediate the situation will be carried out using the remote control consol. The consol will be located in

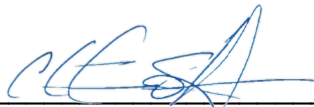
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area removed from the drill in order to ensure that employees are protected from any potential falling objects.

- g) Once the failed rods have been removed and replaced, all remaining rods in the hole will be removed in order to ensure no other rods in the drill string have been damaged.
- h) The site supervisor will determine that the equipment is in the condition required to resume drilling operations & will give the sign off for drilling to resume.
- i) The site supervisor will have the failed rods tagged & returned to the vendor in order to determine the likely cause of the failure.
- j) The site supervisor will be responsible for completing an incident report that will detail the rod failure event, the recovery operation and resulting corrective actions.

34) MRI has equipment adapted to extract broken materials (ex. rods) to keep drilling

Modifications	By	Date
Point # 23: removed the wood bulk head to protect workers during maintenance and bits change and added pull out the V-30 to do this work Point #27: Removed. Was about bulk head	S. Tremblay S. Desrosiers	12-02-2021
Add point #16 of the possibility that the hydraulic hoses can be stuck and the measures to be taken if this occurs.	S. Tremblay	12-01-2023
Added warning on inspections and high-temperature hoses	7-04-2023	S. Tremblay
Addition of a 3-meter distance between the ribbon and the first pieces of equipment in the work area. Addition made to point #7 and on the " STANDARD INSTALLATIONS" following the Amaruk event.	19-12-2023	S. Tremblay S. Desrosiers
Add #23 to push 1 more than the lentgh of the breaktrough hole	21-06-2024	S. Tremblay JF Gagnon



Christian St-Amour
Operations Manager

Standard installations (not scale)
Keep 7 meters between Cubex and booster
(A distance of at least 3 metres (10 feet) must be left between the barricade at the site entrance and the first pieces of equipment (toolbox, oil, etc.).)

